AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

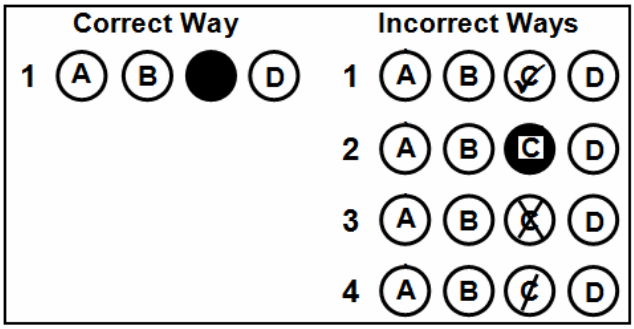
MAY 2014

Chemistry Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. The formula mass of glucose ($C_6H_{12}O_6$) is

   A. 80 amu.
   B. 100 amu.
   C. 180 amu.
   D. 280 amu.

2. In biochemistry, students study about chemical reactions of substances that take place in

   A. the manufacturing of goods.
   B. the body of living organisms.
   C. different parts of the vehicles.
   D. the generation of electrical power.

3. The given equation shows the homolytic breakage of bond between chlorine atoms in the presence of light.

$$Cl_2 \xrightarrow{h\upsilon} 2Cl^{\bullet}$$

   The product ($Cl^{\bullet}$) formed is a/an

   A. atom.
   B. molecule.
   C. free radical.
   D. molecular ion.

4. All of the following conclusions are drawn by Rutherford based on the experiment of scattering alpha particles EXCEPT that

   A. the space inside the atom is hollow.
   B. the central part of atom is positively charged.
   C. the electrons are distributed in space around a nucleus.
   D. the electrons jump to higher energy level by absorbing energy.

5. Which of the following represents the number of sub-atomic particles in carbon – 14 (^{14}C)?

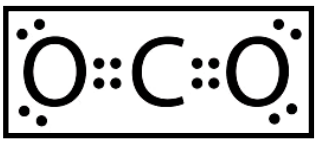
	Proton	**Neutron**	**Electron**
A.	6	6	6
B.	6	7	7
C.	6	8	6
D.	8	6	6

6. Which of the following shows the electronic configuration of a boron atom ($^{11}_{5}B$)?

   A. $1s^2,2s^2,2p^1$
   B. $1s^2,2s^2,2p^2$
   C. $1s^2,2s^2,2p^6$
   D. $1s^2,2s^2,2p^6,3s^1$

7. Which of the following types of bonding exists in the given molecule?

   A. Ionic
   B. Metallic
   C. Covalent
   D. Coordinate covalent



8. All of the following elements on ionization form a cation EXCEPT

   A. silver.
   B. lithium.
   C. oxygen.
   D. calcium.

9. When ammonia (NH_3) reacts with hydrogen ion (H^+), it forms ammonium ion (NH_4^+). A coordinate covalent bond is formed in ammonium ion because

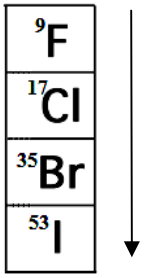
   A. nitrogen gains and hydrogen loses electrons.
   B. nitrogen attains stability by gaining electrons.
   C. nitrogen and hydrogen mutually share an electron pair.
   D. nitrogen provides a lone pair of electrons to hydrogen ion.

10. Which of the following elements exists as a diatomic molecule?

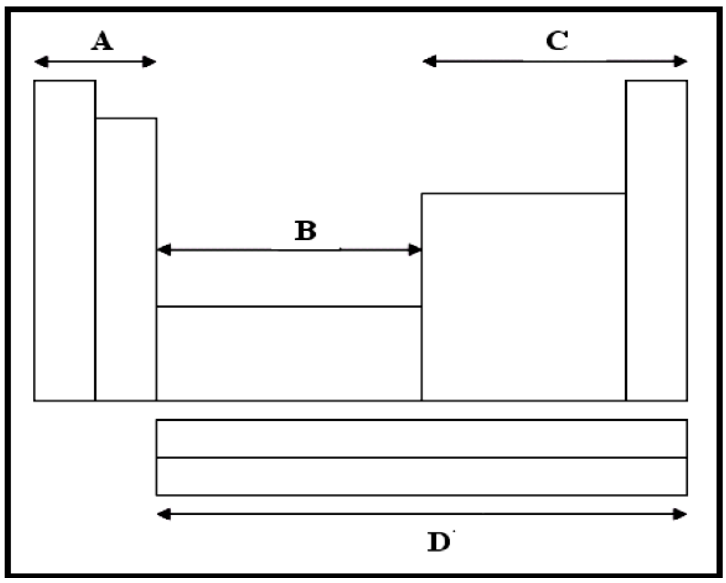
    A. Neon
    B. Sodium
    C. Lithium
    D. Chlorine

11. The given table shows elements of the halogen family. What happens to electron affinity and shielding effect when we move down the group?

	Electron Affinity	Shielding Effect
A.	Decreases	Increases
B.	Increases	Decreases
C.	Decreases	Decreases
D.	Increases	Increases



12. Which of the following represents the d-block elements in the given sketch of the periodic table?



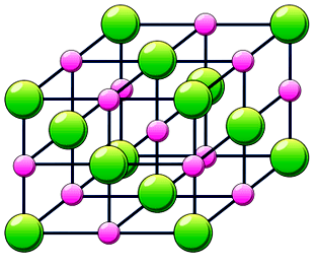
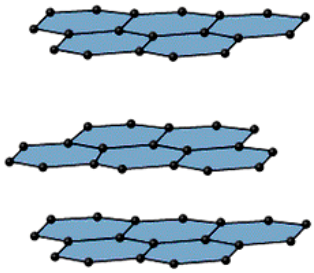
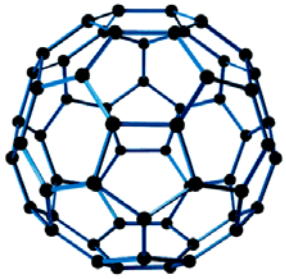
13. Which of the following matter is WRONGLY matched with its property?

	Matter	Property of Particles
A.	Balloon	Can effuse
B.	Table	Can be compressed
C.	Water	Has a definite volume
D.	Smoke	Has weak intermolecular forces

14. Which of the following factors decreases the boiling point of a liquid?

A. Stronger intermolecular forces
B. Lower external pressure
C. Lower vapour pressure
D. Stronger polarity

15. Diamond is an allotropic form of carbon. Which of the following shows the structure of diamond?

	
A.	**B.**
	
C.	**D.**

16. Which of the following information shows that the solution is supersaturated?

Type of Solution	Compound	Temperature	Mass of Solute in 100ml of Water
A.	NH_3	90°C	10g
B.	NH_4Cl	80°C	70g
C.	Na_2SO_4	80°C	20g
D.	$NaNO_3$	25°C	10g

17. Which of the following is NOT an example of a solution made up of liquid into solid?

A. Alloys
B. Butter
C. Cheese
D. Amalgam

18. All of the following substances can dissolve in water EXCEPT

A. salt.
B. sugar.
C. grease.
D. ethanol.

19. Which of the following is the reducing agent in the given reaction?

$$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$$

A. Zn
B. Cu
C. $CuSO_4$
D. $ZnSO_4$

20. All of the following factors speed up the process of corrosion in iron EXCEPT

A. acidic medium.
B. sunlight.
C. oxygen.
D. water.

21. Galvanizing is a method used to prevent the corrosion of iron. This method basically involves the coating of iron with

A. tin.
B. zinc.
C. copper.
D. chromium.

22. Which of the following elements on reaction with oxygen forms a basic oxide?

A. Phosphorus
B. Sulphur
C. Sodium
D. Silicon

23. Platinum alloyed with palladium and rhodium is used

A. in photographic films.
B. in the making of coins.
C. as a catalyst in automobiles.
D. as a coolant in nuclear reactors.

24. Gold is a noble metal due to which it is NOT

A. ductile.
B. malleable.
C. affected by atmosphere.
D. alloyed with other metals.

25. What products are formed when sodium metal reacts with water?

A. Sodium hydride and oxygen gas
B. Sodium oxide and hydrogen gas
C. Sodium hydroxide and oxygen gas
D. Sodium hydroxide and hydrogen gas

END OF PAPER

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